

A FORMAL PHILOSOPHICAL-MATHEMATICAL PROPOSAL

David's Theory of Perceptual Abstraction

Two Mathematical Foundations:
Ontological and Relational

$$H = E - h$$

What remains of a phenomenon after its encounter with abstraction?
Does the possibility of transformation already belong to reality, or
does it arise from the relation between reality and an interpreting
subject?

CONCEPTUAL STATUS

This document develops two alternative formalizations of the same founding equation. The propositions are mathematical consequences of stated assumptions; interpretations concerning perception, collective behavior, or intelligence remain hypotheses requiring operational definitions and empirical tests.

Abstract

David's Theory of Perceptual Abstraction begins from a difference relation, $H = E - h$. In one foundation, both the phenomenal magnitude E and the transformative potential h are intrinsic properties of phenomena; the resulting residual H is observer-independent. In another, E belongs to a phenomenon while h arises from a subject-phenomenon interaction; the resulting residual is relational and may vary across people and time. The first model offers ontological simplicity but requires an additional map to describe diverse subjective perceptions. The second directly accommodates interpretive differences and supports an explicit network model of relational amplification. We state assumptions, establish elementary propositions, compare what each model predicts, and propose a hybrid extension and possible falsification tests. No connection with physical energy or Planck's reduced constant is assumed.

Reader's guide

1. **Common foundations:** what the symbols mean, and what mathematical rigor does and does not establish.
2. **Model A – Ontological:** transformative potential as an intrinsic property of a phenomenon.
3. **Model B – Relational:** transformative potential as a property of interaction and social coupling.
4. **Comparison and research program:** an optional hybrid and tests that could distinguish the models.

THE CENTRAL RELATION

$$H = E - h$$

E – **phenomenal magnitude.** A formally assigned intensity of a phenomenon, not necessarily physical energy.

h – **transformative or abstractive capacity.** Its source is the precise point of disagreement between the two models.

H – **residual.** A signed difference indicating which of the two magnitudes dominates on a shared scale.

Three regimes. $H > 0$ means phenomenal magnitude dominates; $H = 0$ means equality on the chosen scale; $H < 0$ means transformative capacity dominates. These are definitions of regimes, not direct measurements of creativity or well-being.

1. Common mathematical foundations

Assumption 1.1 (Common scale). E and h are nonnegative real-valued quantities expressed on a common measurement scale. Subtraction requires a consistent scale, units and operational procedures.

Definition 1.2 (Signed perceptual residual). For a given admissible pair $(E, h) \in \mathbb{R}_{\geq 0}^2$, define

$$\boxed{H := E - h.} \quad (1)$$

Its magnitude is $D := |H|$; its positive and negative parts are $H_+ := \max(H, 0)$ and $H_- := \max(-H, 0)$.

Proposition 1.3 (Decomposition). For every admissible pair,

$$H = H_+ - H_-, \quad |H| = H_+ + H_-, \quad H_+ H_- = 0. \quad (2)$$

Proof. If $H \geq 0$, then $(H_+, H_-) = (H, 0)$. If $H < 0$, then $(H_+, H_-) = (0, -H)$. Each identity follows in both cases. $\square$

Proposition 1.4 (Monotonicity). Holding h fixed, increasing E increases H by the same amount. Holding E fixed, increasing h decreases H by the same amount.

Proof. For increments ΔE and Δh , the resulting change is $\Delta H = \Delta E - \Delta h$. $\square$

Limits of the formalism. Algebra cannot, by itself, show that an observed mental process follows this model. It also cannot define a unique measurement of E or h . Different meanings for those symbols may produce entirely different empirical theories even though they share exactly the same equation.

Terminology. The phrase *perceptual residual* should be treated as a theoretical name for H . It does not assert the existence of a literal physical residue.

2. Model A: the ontological foundation

TRANSFORMATIVE POTENTIAL IS INTRINSIC TO REALITY

Definition 2.1 (Ontological model). Let $\mathcal{F}$ be a nonempty set of phenomena, with two functions

$$E_A : \mathcal{F} \rightarrow \mathbb{R}_{\geq 0}, \quad h_A : \mathcal{F} \rightarrow \mathbb{R}_{\geq 0}. \quad (3)$$

Define

$$\boxed{H_A(f) = E_A(f) - h_A(f).} \quad (4)$$

In Model A, the phenomenon itself is assigned both its degree of manifestation and its degree of transformability. Neither assignment takes a subject as an argument.

Proposition 2.2 (Observer invariance). If two observers encounter the same phenomenon f under the same model state, its ontological residual is the same for both observers.

Proof. H_A is a function only of f . Observer identity does not appear among its arguments. Therefore $H_A(f \mid s_1) = H_A(f \mid s_2) = H_A(f)$, where the conditional notation records the comparison rather than a new functional dependence. $\square$

2.1 Interpretation and limits of Model A

The three ontological regimes read:

$$H_A(f) > 0 \iff E_A(f) > h_A(f) \quad \text{manifestation dominates,} \quad (5)$$

$$H_A(f) = 0 \iff E_A(f) = h_A(f) \quad \text{structural balance,} \quad (6)$$

$$H_A(f) < 0 \iff E_A(f) < h_A(f) \quad \text{transformability dominates.} \quad (7)$$

Model A does *not* assert that the subject sees precisely H_A . Indeed, subjective variation must be represented using an additional interpretation map.

Definition 2.3 (Subjective reading of an intrinsic residual). For every subject s , let $\Phi_s : \mathbb{R} \rightarrow \mathcal{P}$ map ontological residuals into a space $\mathcal{P}$ of perceived representations. Then

$$P_A(s, f) = \Phi_s(H_A(f)). \quad (8)$$

Different subjects can therefore perceive a single intrinsic residual differently without changing its ontological value. But unless Φ_s is independently constrained, much of the explanatory work moves from H_A into the new function.

A falsifiable consequence, with qualifications. If an operational measure is claimed to estimate H_A itself, the measure should be invariant across observers after fixing the same phenomenon and relevant conditions. Differences in subjects' reports do not alone refute Model A: they may arise through Φ_s . A serious test needs an independent protocol for the claimed intrinsic quantity.

2.2 A philosophical reading

The potential for transformation precedes interpretation. A mind does not bring the possibility of transformation into existence; it discovers, realizes, or expresses possibilities already attributed to the phenomenon. This is an ontological hypothesis, not a theorem obtained from subtraction.

3. Model B: the relational foundation

TRANSFORMATIVE CAPACITY EMERGES IN AN ENCOUNTER

Definition 3.1 (Relational model). Let $\mathcal{S}$ be a nonempty set of subjects, $\mathcal{F}$ a set of phenomena and $\mathcal{T}$ a time domain. Define

$$E_B : \mathcal{F} \times \mathcal{T} \rightarrow \mathbb{R}_{\geq 0}, \quad (9)$$

$$h_B : \mathcal{S} \times \mathcal{F} \times \mathcal{T} \rightarrow \mathbb{R}_{\geq 0}. \quad (10)$$

The relational residual is

$$\boxed{H_B(s, f, t) = E_B(f, t) - h_B(s, f, t)}. \quad (11)$$

Here the subject's participation is constitutive of the model's abstractive term. A phenomenon has no single subject-independent residual of the form H_B unless additional conditions make all the values coincide.

Proposition 3.2 (Difference across subjects). For any two subjects s_1, s_2 and any fixed (f, t) ,

$$H_B(s_1, f, t) - H_B(s_2, f, t) = h_B(s_2, f, t) - h_B(s_1, f, t). \quad (12)$$

Proof. Subtract the two definitions of H_B . The common $E_B(f, t)$ term cancels. $\square$

3.1 Time dependence

Suppose E_B and h_B are differentiable along a fixed subject–phenomenon encounter. Then

$$\boxed{\frac{dH_B}{dt} = \frac{dE_B}{dt} - \frac{dh_B}{dt}}. \quad (13)$$

If the assigned phenomenal magnitude is locally constant,

$$\frac{dE_B}{dt} = 0 \quad \implies \quad \frac{dH_B}{dt} = -\frac{dh_B}{dt}. \quad (14)$$

The result does not establish that people always become more creative over time. It says that *if* the model assigns a rising capacity h_B while holding E_B fixed, the residual decreases at the same rate.

3.2 Worked example: one phenomenon, two readers

Let the phenomenal magnitude be $E_B(f, t) = 10$ on a hypothetical scale. For subjects s_1 and s_2 , suppose $h_B(s_1, f, t) = 3$ and $h_B(s_2, f, t) = 12$.

$$H_B(s_1, f, t) = 7, \quad H_B(s_2, f, t) = -2. \quad (15)$$

The encounter lies in different signed regimes even though its phenomenal input is held fixed. This is a mathematical illustration, not an empirical finding.

4. The relational network extension

FROM INDIVIDUAL ABSTRACTION TO COLLECTIVE COUPLING

To represent interactions among people, take n subjects. Let $b_i \geq 0$ denote a baseline abstractive capacity. Let $W = (w_{ij})$ be a nonnegative, row-stochastic matrix (each row sums to 1). Let $0 \leq \lambda < 1$ describe a uniform coupling strength.

Assumption 4.1 (Linear relational amplification). For a fixed shared phenomenon and time, the capacities satisfy

$$h_i = b_i + \lambda \sum_{j=1}^n w_{ij} h_j, \quad i = 1, \dots, n. \quad (16)$$

The matrix equation is

$$\mathbf{h} = \mathbf{b} + \lambda W \mathbf{h}. \quad (17)$$

Proposition 4.2 (Existence, uniqueness and nonnegativity). Under the assumptions above, there is a unique nonnegative solution

$$\mathbf{h} = (I - \lambda W)^{-1} \mathbf{b} = \sum_{k=0}^{\infty} \lambda^k W^k \mathbf{b}. \quad (18)$$

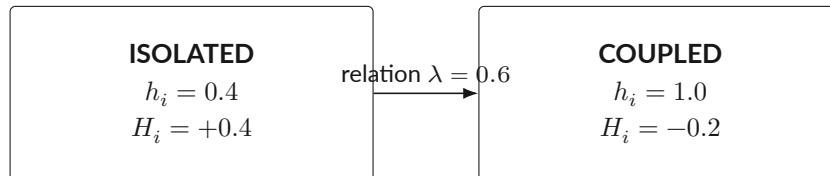
Proof. Because W is row-stochastic, its spectral radius is $\rho(W) = 1$. Hence $\rho(\lambda W) = \lambda < 1$. The Neumann series converges and equals $(I - \lambda W)^{-1}$. Since W , λ and $\mathbf{b}$ are nonnegative, every term of the series is nonnegative. $\square$

4.1 A two-person demonstration

Let

$$\mathbf{b} = \begin{pmatrix} 0.4 \\ 0.4 \end{pmatrix}, \quad W = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \lambda = 0.6, \quad E_B = 0.8. \quad (19)$$

Without coupling ($\lambda = 0$), each individual has $h_i = 0.4$, hence $H_i = +0.4$. With coupling, symmetry gives $h = 0.4 + 0.6h$, and therefore $h = 1$ and $H_i = -0.2$.



The sign change is guaranteed by these chosen values and assumptions, not by evidence that real social interaction always improves human reasoning. In a model with costs, conflict, misinformation or negative influences, amplification may be absent or harmful.

4.2 Why this is not yet a theory of human eusociality

A rise in modeled capacity h_i is not the same as cooperation, reproductive division of labor, or evolutionary fitness. Strict *eusociality* has a biological meaning; for human large-scale cooperation, *ultrasociality* is often a more appropriate label. The network equation can supply a candidate mechanism for *relational amplification*; it cannot by itself explain the evolutionary emergence of human institutions. To connect the two, a distinct performance or payoff measure must be collected. For example, let B_{group} be observed group performance and B_{isolated} a defined independent baseline:

$$\Delta B = B_{\text{group}} - B_{\text{isolated}}. \quad (20)$$

A testable question is whether changes in estimated coupling λ predict changes in ΔB better than established competing explanations. The sign of H *does not logically determine* the sign of ΔB .

5. A comparison of the two foundations

Criterion	Model A: Ontological	Model B: Relational
Origin of E	Phenomenon	Phenomenon
Origin of h	Intrinsic to phenomenon	Subject–phenomenon encounter
Observer-independent H	Yes, by definition	Not generally
Different subject reports	Need an interpretation map Φ_s	Can arise directly from h_B
Role of social links	May alter phenomena or readings	Can enter the defining capacity
Primary strength	Structural simplicity	Direct account of relational variation
Primary limitation	Subject differences are external to H_A	Requires measuring an interaction-dependent h_B

Neither model wins from algebra alone. Both are internally coherent under their definitions. Choosing between them requires specified measurement procedures and distinct empirical predictions. A flexible model with extra free functions can fit more observations without being more explanatory.

6. Optional synthesis: the hybrid hypothesis

The two ideas can be combined without identifying them as the same theory. Decompose an abstractive capacity into an intrinsic contribution and a relational contribution:

$$h_{\text{total}}(s, f, t) = h_0(f, t) + h_R(s, f, t). \quad (21)$$

The resulting residual is

$$H_C(s, f, t) = E(f, t) - h_0(f, t) - h_R(s, f, t). \quad (22)$$

Here h_0 is assigned to the phenomenon, whereas h_R is assigned to the encounter. This preserves $H = E - h$ by setting $h = h_0 + h_R$.

Proposition 6.1 (Non-identifiability without further constraints). Observing E and H_C alone does not uniquely determine h_0 and h_R .

Proof. The observation determines only $h_0 + h_R = E - H_C$. For any admissible number c such that $h_0 + c \geq 0$ and $h_R - c \geq 0$, the pair $(h_0 + c, h_R - c)$ gives the same H_C . $\square$

The hybrid is therefore a *candidate research model*, not an automatic resolution of the difference. Independent observations, interventions, or constraints would be necessary to identify its components.

7. A program of empirical tests

1. **Operationalize the constructs.** Define the phenomenon class, a repeatable metric of E , and an independent method of assessing h . Ensure a common scale before subtraction.
2. **Separate intrinsic and subjective variation.** Present standardized stimuli to multiple subjects; repeat observations across sessions and after controlled interaction.
3. **Intervene on social coupling.** Randomize interaction structure or information access while keeping the task and resources comparable.
4. **Measure outcomes separately.** Evaluate comprehension, novelty, calibration or cooperative payoffs independently of how h is scored.
5. **Compare preregistered predictions.** Test Models A and B, the hybrid, and simpler baselines on held-out observations; account for added parameters.
6. **Report uncertainty.** Distinguish estimated sign changes from changes smaller than measurement error, and publish null as well as positive outcomes.

7.1 Uncertainty and the residual's sign

If observations satisfy $|\widehat{E} - E| \leq \varepsilon_E$ and $|\widehat{h} - h| \leq \varepsilon_h$, then

$$|\widehat{H} - H| \leq \varepsilon_E + \varepsilon_h, \quad (23)$$

by the triangle inequality. Therefore the sign of H is certified by these bounds only when

$$|\widehat{H}| > \varepsilon_E + \varepsilon_h. \quad (24)$$

Inside that band the proposed regime is unresolved under the available error guarantees.

8. Implications for artificial intelligence

The same distinction can be used to frame questions about a large language model, without claiming that a model possesses human perception or consciousness.

Ontological interpretation. An AI system can, in principle, process an assigned phenomenon and its intrinsic modeled properties without a human interacting on every inference step. A subject-specific interpretation map may still be needed to explain different outputs.

Relational interpretation. The information available to an AI, including context, feedback and task-specific human preferences, may alter how it represents and responds to a phenomenon. This gives a possible source of h_B but does not show that human interaction is logically necessary for every computation.

Conditional necessity. If successful behavior depends on a private human preference unavailable from the system's observations, more computational capacity alone cannot recover that exact preference reliably. The missing information must be supplied or inferred from additional evidence. This is an information constraint, not a proof of a universal biological need for humans.

9. Final statement

The ontological foundation holds that a phenomenon already possesses a difference between manifestation and transformability. The relational foundation holds that this difference can instead be constituted in an encounter between the world and an interpreting subject. Both are expressed by the same mathematical form, but they do not make the same claims about reality.

$$H_A(f) = E_A(f) - h_A(f)$$

$$H_B(s, f, t) = E_B(f, t) - h_B(s, f, t)$$

The decisive research question is not which equation looks more elegant, but whether transformative potential can be operationally detected as an invariant property of phenomena, or whether it must be attributed to the dynamics of interaction. Until that question is tested, the theory is a structured philosophical–mathematical proposal, not an established law of perception or social evolution.